

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
7	(a)			It is cheaper than the traditional method ☐ It uses less energy ☒ It reduces carbon dioxide emissions ☒ It uses gold nano-particles ☐ It uses more fuel ☐			2	2		
	(b)			$2\text{CH}_4 + \text{O}_2 \longrightarrow 2\text{CH}_3\text{OH}$ award (1) for reactant and product award (1) for balancing - can only be awarded if reactant and product are correct			2	2		
	(c)			The melting points of gold nano-particles and bulk gold are the same ☐ Gold nano-particles have a fixed melting point value ☐ Smaller gold nano-particles have higher melting points than larger gold nano-particles ☐ The melting point of gold nano-particles depends on their size ☒			1	1		
				Question 7 total	0	0	5	5	0	0